k-Anonymity-Based Data Anonymization and Re-identification Risk Analysis

MSCS 714 Data Privacy Seminar

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Abstract

This report presents the implementation and evaluation of a k-anonymity pipeline for the Adult Census Income dataset. The study compares two quasi-identifier sets, two anonymization strategies, and three privacy levels ($k = 2, 5, 10$), producing 12 experiment settings in total. The implemented pipeline uses only generalization and suppression, as required by the course project. A synthetic auxiliary dataset is then used to perform exact-match linkage attacks against each anonymized release. The main result is that all releases reduce the unique match rate to zero, but they do so in very different ways. The generalization-first strategy keeps all rows but achieves privacy by heavily coarsening the quasi-identifiers, while targeted suppression keeps more original detail but removes between 5.3% and 28.2% of records for QI Set A and between 7.8% and 18.7% for QI Set B. The experiments show a clear privacy–utility trade-off: increasing k increases ambiguity and failed matches for the attacker, but also increases data loss and distributional distortion. Among the tested configurations, targeted suppression with low k provides the best practical balance between privacy protection and utility.

1 Introduction

Removing names from a released dataset does not guarantee privacy. In many realistic data-sharing scenarios, individuals can still be re-identified by combining *quasi-identifiers* such as age, sex, education, occupation, or country with external information. This is the classic linkage-attack problem: even when direct identifiers are removed, a rare combination of quasi-identifiers may still point to a single person.

This project studies that risk through the lens of **k-anonymity**. A dataset satisfies k-anonymity when every released combination of selected quasi-identifiers appears in at least k records. In principle, this should prevent an attacker from uniquely identifying a person by exact matching on the released quasi-identifiers. In practice, however, stronger anonymization often comes at the cost of lower analytical utility.

The goal of this project is therefore twofold:

- reduce re-identification risk through k-anonymity-based transformations, and
- measure how those transformations affect the usefulness of the data.

The implementation follows the exact scope of Project 4 in the seminar: no differential privacy, no cryptography, and no l-diversity extensions. Only generalization and suppression are used.

2 Problem Statement and Objectives

The project investigates the following question:

How do different values of k, different quasi-identifier choices, and different anonymization strategies affect exact-match linkage risk and data utility in the Adult Census Income dataset?

The concrete objectives are:

- build a reproducible Python pipeline for k-anonymization;
- compare $k = 2$, $k = 5$, and $k = 10$;
- evaluate two quasi-identifier (QI) sets with different sparsity profiles;
- compare a generalization-first strategy against a targeted-suppression strategy;
- simulate an adversary using auxiliary data and report unique, ambiguous, and failed matches;
- measure utility loss with interpretable summary metrics and plots.

3 Related Work

k-anonymity was formalized as a model for privacy-preserving data release by Samarati and Sweeney and later developed further by Sweeney, who demonstrated that combinations of seemingly innocuous attributes can still enable re-identification after explicit identifiers are removed [1, 2]. This insight directly motivates the threat model used in this project.

Bayardo and Agrawal studied optimal k-anonymization under generalization and suppression, showing that privacy enforcement and utility preservation must be balanced carefully [3]. Their work motivates the heuristic strategies implemented here rather than an expensive global optimization approach.

Researchers later proposed *l-diversity* and *t-closeness* to address cases where k-anonymity alone still leaves sensitive attributes vulnerable to inference [4, 5]. These extensions are outside the scope of the course assignment, but they are important when interpreting the limitations of the results.

For a broader overview, Fung et al. survey anonymization models, attack assumptions, and utility trade-offs in privacy-preserving data publishing [6]. The Adult dataset remains a standard benchmark for experiments in privacy and tabular data analysis because of its realistic mixture of continuous and categorical demographic attributes [7, 8].

4 Dataset Description

4.1 Source and Size

The project uses the **Adult Census Income** dataset from the UCI Machine Learning Repository. The local training split used in this project is stored at `data/raw/adult/adult.data`. It contains **32,561** records and **15** columns, which comfortably satisfies the minimum data-size expectation in the course description.

4.2 Observed Data Characteristics

Inspection of the local file produced the following summary:

- total rows: 32,561
- total columns: 15
- age range: 17 to 90, mean 38.58

- hours-per-week range: 1 to 99, mean 40.44
- income distribution: 24,720 records with `<=50K` and 7,841 records with `>50K`
- missing values encoded as `?`: `workclass` (1,836), `occupation` (1,843), and `native-country` (583)

The dataset is well suited for this project because it includes both common and sparse attribute values, making it possible to study the tension between privacy protection and information loss.

4.3 Attribute Roles

Table 1 summarizes how the columns were used in the experiments.

Attribute	Role	Rationale
`income`	Sensitive attribute	Preserved for privacy and utility evaluation.
`age`, `sex`, `education`, `marital-status`	QI Set A	Moderate-risk demographic combination with no missing values.
`age`, `sex`, `occupation`, `native-country`	QI Set B	More detailed, sparser combination that represents a stronger linkage scenario.
`hours-per-week`, `education-num`	Utility attributes	Kept to measure summary-statistic distortion.
`workclass`, `relationship`, `race`, `capital-gain`, `capital-loss`, `fnlwgt`	Supporting / unused in matching	Available in the release, but not part of the chosen QI sets.

Table 1: Attribute roles used in the project.

4.4 Baseline Risk Before Anonymization

Before any transformation, the raw data already shows clear re-identification exposure:

- QI Set A yields 4,331 equivalence classes, including 1,724 singleton classes.
- QI Set B yields 4,308 equivalence classes, including 2,526 singleton classes.

These singleton classes are exactly the patterns that make record linkage dangerous in practice.

5 Threat Model

The attacker is assumed to possess an auxiliary dataset that contains quasi-identifiers for a subset of individuals plus an internal `person_id`. The attacker does not know the sensitive attribute (`income`) but can observe the anonymized release and attempt exact matching on the visible quasi-identifiers.

In the implementation, the auxiliary dataset is created by sampling 20% of the original records, producing **6,512** auxiliary records. This is a controlled simulation rather than an external real-world database, but it provides a direct way to measure how often exact linkage remains possible after anonymization.

6 Methodology

6.1 Preprocessing

The pipeline loads the Adult dataset with explicit column names, trims categorical whitespace, converts `?` to missing values, and fills missing QI-related categories with `Unknown`. A synthetic `person_id` is added only for evaluation. The sensitive attribute is preserved as both the original binary label and a numeric indicator column.

6.2 Quasi-Identifier Sets

The experiments use the following two QI sets:

- **QI Set A:** {`age`, `sex`, `education`, `marital-status`}
- **QI Set B:** {`age`, `sex`, `occupation`, `native-country`}

QI Set B is intentionally more difficult because occupation and country are more detailed and contain rare values.

6.3 Generalization Hierarchies

The pipeline applies interpretable hierarchies rather than opaque transformations:

- **Age:** exact age → 5-year bins → 10-year bins → broad age groups → full suppression
- **Education:** exact label → grouped schooling levels → pre-college / college-or-higher → full suppression
- **Marital status:** exact label → simplified marital groups → partnered / not-partnered → full suppression
- **Occupation:** exact label → job family → white-collar / blue-collar / service / other → full suppression
- **Native country:** exact country → region → US / non-US → full suppression
- **Sex:** exact value → full suppression

6.4 Anonymization Strategies

Two strategies were implemented:

Generalization-first This strategy progressively generalizes quasi-identifiers until all equivalence classes meet the k-anonymity threshold. It prioritizes row retention and suppresses rows only as a last resort. In the experiments, this strategy never removed rows.

Targeted suppression This strategy uses little or no generalization and instead removes rows that remain too rare under the chosen QI set. It preserves more original categorical detail in retained rows, but reduces release size.

6.5 Risk and Utility Metrics

For each release, the following metrics were computed:

- **Unique match rate**
- **Ambiguous match rate**
- **No match rate**
- **Retention rate** and **suppression rate**
- **Income distribution shift**
- **Absolute change in mean and variance of hours per week**

7 Experimental Design

The full experiment grid exactly matches the course requirements:

$$3 \text{ k-values} \times 2 \text{ QI sets} \times 2 \text{ strategies} = 12 \text{ runs}.$$

The evaluated settings were:

- $k \in \{2, 5, 10\}$
- QI Set A and QI Set B
- generalization-first and targeted suppression

All experiment outputs were saved to:

- `results/metrics/experiment_summary.csv`
- `results/releases/*.csv`
- `results/figures/risk_vs_k.png`
- `results/figures/utility_vs_k.png`

8 Results

8.1 Main Finding

The strongest headline result is simple: **the unique match rate was reduced to zero in all 12 experiment settings**. This is exactly what we expect from a correctly enforced exact-match k-anonymity release. However, the mechanism by which that happened differed sharply across strategies:

- **Generalization-first** achieved zero unique matches by aggressively coarsening quasi-identifiers, often to the point of full suppression.
- **Targeted suppression** achieved zero unique matches by keeping the original QI values for retained rows and removing rare records instead.

8.2 Risk Trends

Figure 1 plots re-identification risk as a function of k. Because unique matches are eliminated in every release, the meaningful distinction is between *ambiguous matches* and *no matches*. Generalization-first forces all auxiliary records into ambiguous matches, while targeted suppression increasingly converts risky individuals into no-match cases as k grows.

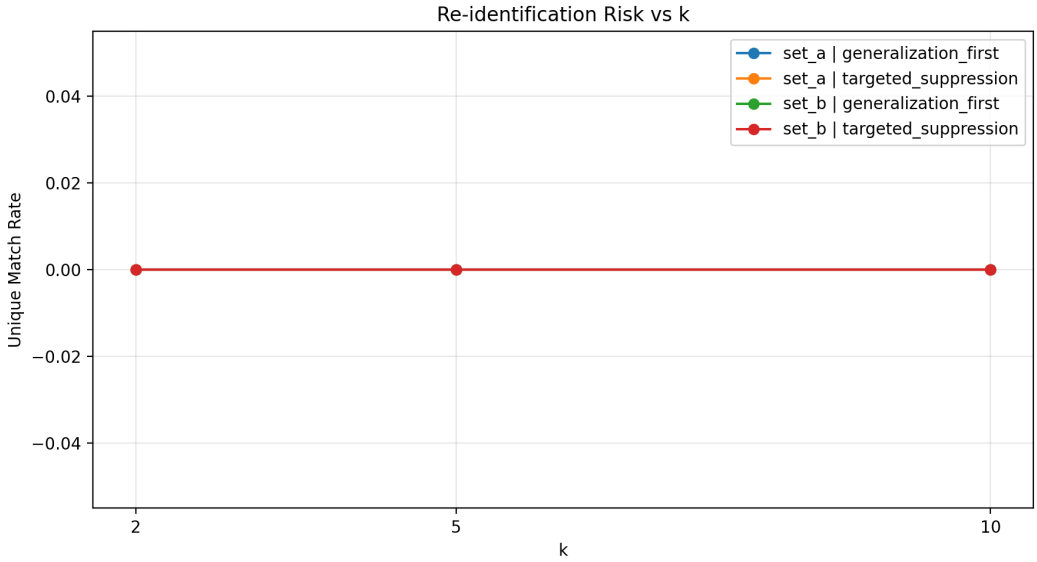

Figure 1: Re-identification risk versus k for both QI sets and both anonymization strategies.

8.3 Utility Trends

Figure 2 shows utility loss measured by income-distribution shift. Targeted suppression produces a visible increase in distortion as k rises. In contrast, generalization-first appears to have almost zero distributional distortion under this metric because it does not remove rows. That does **not** mean it preserves utility better overall; it means this specific metric does not capture how much semantic detail is lost when age, sex, country, or education are generalized away.

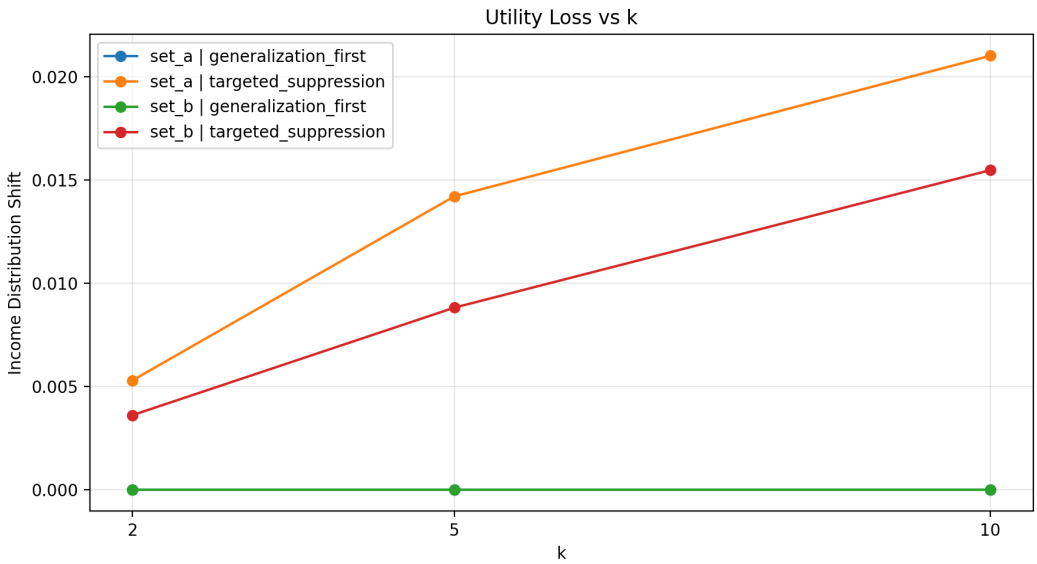

Figure 2: Utility loss versus k, using income-distribution shift as the plotted metric.

8.4 Targeted Suppression Results

Table 2 summarizes the most interpretable strategy from a practical release perspective. For both QI sets, higher k leads to more removed rows, a larger no-match rate for the attacker, and larger utility distortion.

QI Set	k	Retention	Suppression	No-Match	Income Shift
Set A	2	94.7%	5.3%	5.2%	0.0053
Set A	5	83.8%	16.2%	16.7%	0.0142
Set A	10	71.8%	28.2%	29.0%	0.0210
Set B	2	92.2%	7.8%	7.6%	0.0036
Set B	5	86.9%	13.1%	12.6%	0.0088
Set B	10	81.3%	18.7%	18.1%	0.0155

Table 2: Targeted-suppression results across both quasi-identifier sets.

Several patterns are clear:

- For both QI sets, stronger privacy levels increase suppression and no-match rates.
- QI Set A is slightly easier to preserve at low k, but becomes more distorted than QI Set B at higher k.
- The best overall retention is achieved by QI Set A with targeted suppression at $k = 2$ (94.7% retained rows).
- The smallest income-distribution shift is achieved by QI Set B with targeted suppression at $k = 2$ (0.0036).

8.5 Generalization-First Results

Table 3 shows the generalization patterns selected by the generalization-first strategy. This strategy retained all 32,561 rows in every run and made every auxiliary match ambiguous, but it did so by using very coarse released quasi-identifiers.

QI Set	k	Generalization Levels Applied
Set A	2	`age:4, sex:1, education:2, marital_status:0`
Set A	5	`age:4, sex:1, education:3, marital_status:0`
Set A	10	`age:4, sex:1, education:3, marital_status:0`
Set B	2	`age:4, sex:1, occupation:0, native_country:3`
Set B	5	`age:4, sex:1, occupation:0, native_country:3`
Set B	10	`age:4, sex:1, occupation:2, native_country:3`

Table 3: Generalization levels selected by the generalization-first strategy.

These settings show why row-retention alone is not a sufficient utility metric. For example, the strategy frequently suppresses `sex` entirely and pushes `age` to its maximum generalization level. That preserves row count, but substantially reduces the analytical value of the released quasi-identifiers.

9 Discussion

The experiments confirm the central privacy–utility trade-off expected from k-anonymity:

- stronger privacy requirements reduce the attacker's ability to isolate records;
- but stronger privacy also either removes more records or coarsens the data more aggressively.

The exact-match attack outcome is especially instructive. Since every release satisfies k-anonymity under the chosen QIs, unique matches disappear completely. Therefore, the more meaningful post-anonymization distinction becomes:

- **ambiguous match**: the attacker still finds several possible candidates;
- **no match**: the auxiliary individual is no longer represented in a directly matchable way.

From a practical point of view, targeted suppression is easier to interpret. It produces a visible and intuitive trade-off curve: as k increases, privacy improves because more risky records disappear from exact matching, but utility decreases because fewer rows remain. Generalization-first produces a different kind of trade-off: it keeps all rows, yet the released QIs become so coarse that much of their value is lost even though simple distribution metrics remain stable.

This also highlights a limitation of the current utility evaluation. Distributional stability alone does not capture the loss of granularity in categorical variables. A release with zero row loss can still be analytically weak if most quasi-identifiers are generalized to broad categories or completely suppressed. For that reason, the results should be interpreted jointly with the recorded generalization levels, not only with the utility plot.

10 Limitations and Ethical Considerations

This project intentionally evaluates only **k-anonymity**, because that is the assigned project scope. As a result, the releases are not protected against all forms of attribute disclosure. If an equivalence class becomes homogeneous in income, an attacker may still infer the sensitive attribute even without uniquely identifying a person.

The attacker model is also simplified: the auxiliary dataset is sampled from the same underlying data rather than obtained from an independent real-world source. This makes the evaluation controlled and reproducible, but it does not fully capture every possible linkage environment.

Another limitation is metric selection. The current utility metrics capture row retention and some distributional changes, but they understate semantic information loss when attributes are heavily generalized. A stronger final version of the project could add category-level divergence measures or lightweight predictive utility tests.

Finally, demographic datasets include socially sensitive information, so the analysis should be understood as a study of privacy methodology rather than a statement about the people represented in the data.

11 Reproducibility

The project is fully reproducible from the repository. The core files are:

- `data/raw/adult/adult.data`
- `scripts/run_experiments.py`
- `src/k_anonymity_risk_analysis/*.py`
- `results/metrics/experiment_summary.csv`

The experiments can be rerun with:

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
python scripts/run_experiments.py
```

12 Conclusion

This project implemented and evaluated a complete k-anonymity pipeline for the Adult Census Income dataset. Across 12 experiment settings, the pipeline consistently removed exact unique matches under the chosen quasi-identifiers. However, the experiments also show that "achieving k-anonymity" does not have a single meaning in practice. One strategy can keep every row by heavily generalizing the released attributes, while another can preserve more original detail by removing risky rows.

Among the tested configurations, low-k targeted suppression provides the strongest practical balance for this project. In particular, targeted suppression with $k = 2$ preserves more than 92% of rows in both QI sets while keeping income distortion low and eliminating unique matches. At the same time, the results make clear that higher k values impose real utility costs, and that utility should be evaluated not only through row counts and distributions, but also through the interpretability of the released quasi-identifiers.

References

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